Answer: A

3 $s_1 = \frac{1}{2} \times a_1 t_1^2 = \frac{1}{2} \times a_1 \times 9$

$s_2 = \frac{1}{2} \times a_2 t_1^2 = \frac{1}{2} \times a_2 \times 9$

$s_2 - s_1 = \frac{1}{2} \times (a_2 - a_1) \times 9 = 12 \quad \ldots (1)$

$s_2 - s_1 = \frac{1}{2} \times (a_2 - a_1) \times 36 \quad \ldots\ldots\ldots(2)$

$\frac{\text{eqn}(2)}{\text{eqn}(1)} = \frac{s_2 - s_1}{12} = \frac{36}{9} = 4$

$s_2 - s_1 = 48$ m

Answer: D

4 Resultant force along GX = $2\cos 60^0 \times 40 = 40$N